Consistent theories do not prove their Godel sentence

The Open Logic Project contributors

openlogic:incompleteness/incompleteness-provability:local-snapshot-20261013; CC BY 4.0

Editorial note. The counted statement is the author lemma labelled lem:cons-G-unprov (484-487); the proof span 489-502 is the author proof environment that immediately follows it.

Editorial note. External reference numbers render the unit-internal labels with hyphens in place of colons (pipeline number charset); the Source reference lines below the body give each label's exact source span.

Editorial scope. One original result of the Open Logic Project unit incompleteness/incompleteness-provability, counted once: consistent theories do not prove their Godel sentence. Only the counted lemma and its own author proof are retained; the source is reproduced without mathematical changes and keeps the CC BY 4.0 licence and the author attribution.

Allowed standard prerequisites (editorial).

- The Godel sentence and its Q-form. The negation of the Godel sentence is equivalent in Q to the existential 'there is x with Prf_T(x, gG_T)' (the unit's equation eqn:qpf).. It converts the derived existential into a derivation of the negated Godel sentence (496-498).
- Representability of the proof predicate in Q. When a derivation of a sentence exists, Q derives the corresponding numeral instance of the proof predicate (the unit's representability standard for Prf_T).. It lifts the concrete witness m to a Q-derivation of OPrf_T(m-bar, gG_T) (491-494).
- Monotonicity of derivability. Whatever Q derives, every extension of Q derives as well.. It carries the Q-refutation of the Godel sentence up to T (498-500).

Source reference. Original equation eqn-qpf: sources/olp-incompleteness-incompleteness-provability/pinned-original-source.tex, source lines 478–478. The author proof cites the unit's equation giving the Q-equivalent existential form of the negated Godel sentence. It is a navigation pointer to the displayed equivalence of the same unit; the counted proof retains its own argument in full.

Lemma 0.1. *If* ThT *is a consistent, !!axiomatizable theory extending* ThQ*, then* Th$T \vdash /!G_{\mathrm{Th}}T$.

Proof. Suppose ThT !!derives $!G_{\mathrm{Th}}T$. Then there *is* !!aderivation, and so, for some number m, the relation Prf[ThT]$(m, \mathrm{gn}\,!G_{\mathrm{Th}}T)$ holds. But then ThQ !!derives the sentence Prf[ThT]$(\underline{m}, \mathrm{gn}\,!G_{\mathrm{Th}}T)$. So Th$Q$!!derives $\exists[x][\mathrm{Prf}[\mathrm{Th}T](x, \mathrm{gn}\,!G_{\mathrm{Th}}T)]$, which is, by definition, Prov[ThT]$(\mathrm{gn}\,!G_{\mathrm{Th}}T)$. By eqn-qpf, Th$Q$!!derives $\neg!G_{\mathrm{Th}}T$, and since ThT extends ThQ, so does ThT. We have shown that if ThT !!derives $!G_{\mathrm{Th}}T$, then it also !!derives $\neg!G_{\mathrm{Th}}T$, and hence it would be inconsistent. □

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